

Post-quantum E2EE overlay for interbank messaging and customer communications in retail banking, leveraging proven overlay approaches from major messaging platforms.

Finance, Banking and Insurance × Data sovereignty and protection × Post-quantum cryptography · OS139 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
60 is the world moving	59 can we play, can we win	MEDIUM 0 named in the evidence	€1.3bn partial evidence · per year	NEXT pilots published no volume pr...	L0 6 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 4.0 published reference(s) in Finance, Banking and Insurance, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is a post-quantum end-to-end encryption (E2EE) overlay for interbank messaging and customer communications in retail banking. It targets banks that need to protect sensitive financial data against future quantum threats, leveraging proven overlay approaches from major messaging platforms. The time is now because post-quantum standards exist and early pilots are emerging, but no volume procurement has occurred yet.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€1.6bn €102m – €4.5bn	€1.3bn €83.5m – €3.6bn	€46.0m €2.9m – €127m	partial
Observed: tendered contract value, annualised	€103m €103m – €196m	€103m €103m – €196m	€3.6m €3.6m – €6.9m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Finance, Banking and Insurance (EU27_2020)	11,116 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Post-quantum cryptography	39.7 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cisce_ran2 · E_SECMDENC	2024
Annual value of one engagement	€1.7m per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-06
Size-weighted buyer base	2,426 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	3.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. ICT security measures by NACE Rev. 2 activity series E_SECMDENC is used as a proxy for Post-quantum cryptography: Encryption users are the migration population. Part of this vertical is not covered separately by the enterprise ICT survey, so the all-activities rate was used for those slices. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. Sector adoption is read from Eurostat's all-activities aggregate for part of this vertical, which the enterprise ICT survey does not cover separately: PROXY — ICT survey excludes the financial sector; PROXY — auxiliary financial activities; PROXY — insurance and pension funding

Observed: tendered contract value, annualised

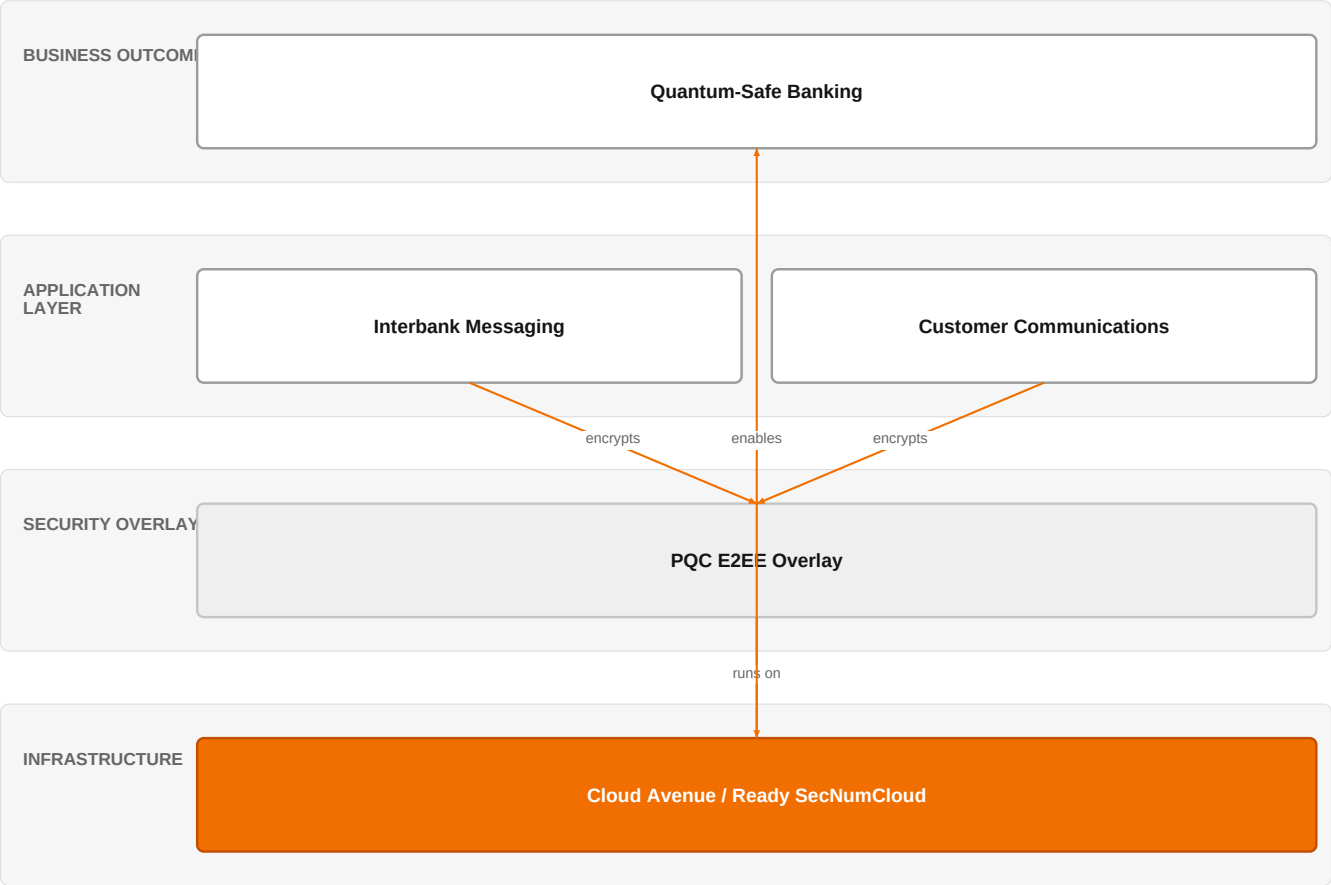
Factor	Value	Basis	Source	As of
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Tendered contract value in matching CPV categories	€103m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-06
Assumed obtainable share of the serviceable market	3.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 19 matching notices, matched on use case via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Post-Quantum E2EE Overlay for Banking



Orange asset Partner Customer-owned Third party / to be sourced

This diagram shows how a post-quantum E2EE overlay sits between banking applications and a sovereign cloud, enabling quantum-safe communications.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Real-time payment gateways still rely on RSA and elliptic curve cryptography, which become vulnerable once large-scale quantum computers can run Shor's algorithm. Post-quantum cryptography (PQC) standards now exist for key exchange and digital signatures, but the urgent question is how to operationalise PQC in complex environments with confidence. A post-quantum E2EE layer has already been developed for major messaging platforms, indicating a market move towards quantum-resistant encryption. Financial institutions are upgrading their cryptographic infrastructure to become quantum safe before large-scale quantum computers materialise.

Sources: SIG-969C5FDB15DD Institute of Electrica 2026-02-02; SIG-178A2ADDCEEE arxiv.org 2026-06-15; SIG-4B7C038778C3 arxiv.org 2026-05-18; SIG-D352790E1A52 useekko.app 2026-07-21

Who buys, and why now

The chief information security officer (CISO) or head of cryptography at a retail bank signs the contract, while the head of payments or head of digital channels feels the operational pain of legacy cryptographic dependencies. The trigger is a regulatory mandate, a quantum-readiness audit, or a high-profile breach in the sector that forces a proactive response. The pain is concrete: existing TLS and messaging encryption are not quantum-safe, and migrating without disrupting

real-time payment latency is a complex, high-risk project.

Sources: SIG-969C5FDB15DD Institute of Electrica 2026-02-02; SIG-178A2ADDCEEE arxiv.org 2026-06-15; SIG-4B7C038778C3 arxiv.org 2026-05-18

Why it is hot now — every claim bound to a dated source

- A post-quantum E2EE layer has been developed for major messaging platforms, indicating a market move towards quantum-resistant encryption. [SIG-D352790E1A52]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a team of cybersecurity experts and Orange Group researchers to conduct a quantum-readiness assessment of the bank's messaging and payment infrastructure. The engagement would then design and deploy a post-quantum E2EE overlay, leveraging the proven overlay approach from major messaging platforms, and run it on Cloud Avenue / Ready SecNumCloud for sovereign, certified hosting. The overlay would be integrated with the bank's existing systems, with the option to use Quantum Defender / quantum-safe offers for additional protection.

Why Orange can win

Orange brings a unique combination of sovereign cloud infrastructure (Cloud Avenue / Ready SecNumCloud) with CISPE certification, which is critical for regulated financial data. The cybersecurity experts and Orange Group researchers provide the deep technical capability to design and deploy PQC overlays. The reference customers Banqsoft and Stø demonstrate experience in financial and public sectors, though the assets are thin on specific PQC deployment references.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Quantum Defender / quantum-safe offers	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
Cloud Avenue / Ready SecNumCloud	offer	L1	offer addresses the use case but not this technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SOC 2 Type II	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
CISPE	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Banqsoft	reference	SUP	published reference in this vertical for this use case	unconfirmed
Stø	reference	SUP	published reference in this vertical for this use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 6.4; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include global systems integrators like Atos / Eviden, IBM, and Accenture, who sell post-quantum cryptography and have strong finance vertical expertise. Security specialists like Airbus Protect, Entrust, ID Quantique, and Thales also sell PQC solutions, with Thales active in finance. The customer will compare Orange's sovereign cloud and certification against these players' broader integration capabilities and specialised cryptographic products.

Sources: SIG-969C5FDB15DD Institute of Electrica 2026-02-02; SIG-178A2ADDCEEE arxiv.org 2026-06-15

Competitor	Category	Position here	Basis	Evidence
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Atos / Eviden	Global systems integrator	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
IBM	Global systems integrator	sells Post-quantum cryptography; active in Finance, Banking and Insurance; competes in Cybersecurity	structural	—
Airbus Protect	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Entrust	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
ID Quantique	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Thales	Security specialist	sells Post-quantum cryptography; active in Finance, Banking and Insurance; competes in Cybersecurity	structural	—
Accenture	Global systems integrator	active in Finance, Banking and Insurance	structural	—
Capgemini	Global systems integrator	active in Finance, Banking and Insurance	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current cryptographic inventory for interbank messaging and customer communication channels?
- 2 Have you conducted a quantum-readiness assessment or received any regulatory guidance on PQC migration?
- 3 What are your latency and availability targets for real-time payment gateways, and how much overhead can you tolerate?
- 4 Do you have a preferred cloud provider or sovereignty requirement for handling financial data?
- 5 Have you evaluated any PQC solutions or pilots from other vendors? What were the outcomes?

Objections you will meet

They will say	You can answer
Quantum computers are not a near-term threat; why invest now?	It's true that large-scale quantum computers are not yet operational, but the migration is complex and time-consuming. Standards exist, and early adopters are moving to avoid a costly rush later. Starting now with a pilot can position you ahead of regulatory pressure.
We already have TLS and encryption in place; why change?	Your current TLS and encryption rely on RSA and elliptic curve cryptography, which are vulnerable to quantum attacks. A post-quantum overlay adds a layer of protection without requiring a complete replacement of your existing infrastructure, minimising disruption.
We are concerned about latency impact on real-time payments.	Latency is a valid concern, and that's why we propose a phased approach. We start with an assessment and pilot on non-critical channels to measure impact, then scale only after we've validated performance against your targets.

Risks and unknowns

The main risk is that the market is still in pilot phase with no volume procurement, so the bank may delay investment. The operational impact of PQC on real-time payment latency is not fully known, and the overlay approach for messaging platforms may not scale to interbank messaging. There is also uncertainty about regulatory timelines for quantum readiness.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a retail bank, incorporating Cloud Avenue / Ready SecNumCloud and Quantum Defender / quantum-safe offers, and referencing the proven overlay approach from major messaging platforms, while highlighting Orange's ISO 27001, SOC 2 Type II, and CISPE certifications.
Sales	In a conversation with a retail bank's head of security, mention that Orange has developed a post-quantum E2EE layer for major messaging platforms and can bring that expertise to their interbank messaging and customer communications.

Strategist / Innovator	Commission a deep-dive brief on post-quantum E2EE overlay for interbank messaging, leveraging Orange Group researchers and cybersecurity experts to validate the technical feasibility and market readiness based on the published pilots.
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Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-969C5FDB15DD	2026-02-02	Institute of Electrical a...	1	Post-Quantum Cryptography Migration Framework for Real-Time Payment Gateways
SIG-6CA0BD99D256	2026-01-08	2026 Second International...	1	Post-Quantum Cryptography Migration Framework for Real-Time Payment Gateways
SIG-228784E54FF2	2026-06-28	FinanceFeeds	2	Top 5 Post-Quantum Cryptography (PQC) Integration Solutions for Enterprise Crypto Wallets - Fin...
SIG-D352790E1A52	2026-07-21	useekko.app	3	Show HN: Ekko – post-quantum E2EE layer for Instagram, WhatsApp, Telegram Web
SIG-178A2ADDCEEE	2026-06-15	arxiv.org	3	Measurement Study of Post-Quantum Readiness of Internet: 2026
SIG-4B7C038778C3	2026-05-18	arxiv.org	3	Operationalising Post Quantum TLS Automated Configuration Profiling and Hybrid PQC Deployment i...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:21+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.